

Evaluate each limit.

5) $\lim_{x \rightarrow -3^-} \frac{2x}{x+3}$

6) $\lim_{x \rightarrow -2^+} \frac{1}{x^2 - 4}$

7) $\lim_{x \rightarrow 3^-} -\frac{4x}{x-3}$

8) $\lim_{x \rightarrow 1} -\frac{3}{x-1}$

9) $\lim_{x \rightarrow -2^-} \frac{x+2}{x^2+x-2}$

10) $\lim_{x \rightarrow -3^-} -\frac{2}{x+3}$

11) $\lim_{x \rightarrow \frac{\pi}{4}^-} 2\sec(2x)$

12) $\lim_{x \rightarrow \frac{3\pi}{4}^+} 2\tan(2x)$

Critical thinking questions:

13) Give an example of a right-sided limit that goes to ∞ as x goes to 5.

14) Give an example of a left-sided limit that goes to ∞ as x goes to 5.